

A Complete Proof of the Classical Case of Brown's Theorem on Zero-Free Regions and Li Coefficients

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Abstract

Let ζ be the Riemann zeta function and $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n]$ its Li coefficients. Brown (2005) proved that the non-negativity of finitely many λ_k yields a zero-free region, and attempted the converse; the proof of the converse depends on a lemma that contains two errors, one of which invalidates it, so that the converse has remained a conjecture (Droll, 2012, Conjecture 1.7.10). We prove the conjecture in the classical case $\tau = 1$ for the zeta function, and with a strictly stronger range: our inequality holds for *all* $k \geq 2$ and all $H > e$, whereas the conjecture asserts the case $k \leq 2H^2 \log H$. The proof is elementary. Its decisive input turns out to be the *negative* sign of the linear coefficient of the zero-counting function, which produces a margin of exactly $\frac{2}{3}|b|H^{-3}$; this is consistent with, and explains, the case distinction $b \geq 0/b < 0$ in the statement of the conjecture.

1 Introduction

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ be the completed zeta function. For $n \geq 1$ the *Li coefficients* are

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho} \right)^n \right],$$

the sum being over the non-trivial zeros. Li's criterion [4] states that the Riemann hypothesis is equivalent to $\lambda_n \geq 0$ for all $n \geq 1$.

Brown [1] considered the quantitative converse: does a zero-free region imply the non-negativity of finitely many λ_k ? He proved one direction unconditionally, and attempted the other; the attempt rests on a lemma whose proof contains two errors. Droll [2] identifies both, resolves one, observes that the other invalidates the proof as it stands, restates the theorem as Conjecture 1.7.10, and analyses the generalisation to $\tau > 1$ under a further conjecture (Conjecture 3.2.7). Palojärvi [3] treats the τ -version with explicit conditions.

Let a, b, c, d be the constants of the counting hypothesis

$$N(T) = aT \log T + bT + \varepsilon(T), \quad |\varepsilon(T)| \leq c \log^+ T + d, \quad a, c, d > 0, \quad 3a + b > 0, \quad (1)$$

where $N(T)$ counts the zeros with $0 < \text{Im } \rho \leq T$. For ζ one has

$$a = \frac{1}{2\pi}, \quad b = -\frac{1 + \log 2\pi}{2\pi}, \quad (2)$$

so that $b < 0$ and $3a + b = 0.0258 \dots > 0$.

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Theorem 1. Let $k \geq 2$ be an integer and $H > e$ a real number. Put $\gamma_\rho = \text{Im } \rho$ and $r_H = (1 + H^{-2})^{1/2}$. Then

$$\sum_{|\gamma_\rho| > H} \left[\left(1 + \frac{1}{\gamma_\rho^2}\right)^{k/2} + \left(1 + \frac{1}{\gamma_\rho^2}\right)^{-k/2} - 2 \right] \leq 2(r_H^k + r_H^{-k} - 2) \left[\frac{a}{3} H \log H + \frac{4a}{9} H + 2c \log H + 2d + \frac{c}{4} \right].$$

A companion note [7] obtains outright positivity of the coefficients for a range linear in the verified height by an elementary phase-window argument that bypasses the defective implication; the present note is complementary, in that it supplies the input inequality of the classical converse for *all* orders. The two share one identity, namely $|1 - 1/\rho|^2 = 1 + (1 - 2\beta)/(\beta^2 + \gamma^2)$ for a zero $\rho = \beta + i\gamma$.

Theorem 1 is the *first* of the two inequalities in Conjecture 3.2.7 of [2], in the classical case ($\tau = 1$), for all $k \geq 2$. (The restriction $k \leq 2T^2 \log T$ belongs to Conjecture 1.7.10 of [2] and does not constrain Conjecture 3.2.7 itself.) Conjecture 3.2.7 contains a *second* inequality (with the factor $9/4$ and the term $b^+ H$); the present note addresses only the first. This inequality is precisely the statement that Droll identifies as the repair of the defective step: his Conjecture 3.2.7, "combined with the previous lemma, provides a modified, generalized version of [Br05, Lemma 5]". Since the conjecture requires only $1 \leq \tau < 2$, the classical case $\tau = 1$ is covered by Theorem 1, and because Brown's remaining inputs are unaffected, the classical case of the theorem follows for all orders. We prove the conjecture modulo two constant-level items (§7), not unconditionally.

2 The summand

Write $u = \gamma^{-2}$ and $x = 1 + u$. We shall use three equivalent forms of the summand:

$$x^{k/2} + x^{-k/2} - 2 = (x^{k/4} - x^{-k/4})^2 = \frac{((1+u)^{k/2} - 1)^2}{(1+u)^{k/2}} = 4 \sinh^2\left(\frac{1}{2} \cdot \frac{k}{2} \log(1+u)\right). \quad (3)$$

The first is a square and hence non-negative; the third is the form used below, being both exact and numerically stable. It was verified to 41 digits.

Lemma 2 (monotonicity). $F(x) := x^{k/2} + x^{-k/2} - 2$ is strictly increasing for $x > 1$.

Proof. $F'(x) = \frac{k}{2}(x^{k/2-1} - x^{-k/2-1})$, which is positive exactly when $x^k > 1$. $\square$

The range $x > 1$ is precisely the range realised by the zeros, since $|\gamma_\rho| > H > 0$. This deserves emphasis: the first of Brown's two errors consists in asserting the monotonicity of $x^k + x^{-k}$ for *all* $x > 0$, where it fails; as Droll notes, that error is "essentially irrelevant in the special case that Brown is addressing", i.e. exactly in the classical case where $x > 1$.

3 Two bounds on the sum

Split

$$\text{LHS} = \text{near} + \text{far}, \quad \text{near} = \sum_{H < |\gamma_\rho| \leq 2H}, \quad \text{far} = \sum_{|\gamma_\rho| > 2H}.$$

Lemma 3 (far). $F(1+u) \leq \frac{k^2}{4} u^2 (1+u)^{(k-4)/2} \leq \frac{k^2}{4} r_H^{(k-4)^+} u^2$, hence $\text{far} \leq \frac{k^2}{4} r_H^{(k-4)^+} S_4(2H)$ with $S_4(X) = \sum_{|\gamma_\rho| > X} \gamma_\rho^{-4}$.

Lemma 4 (near, explicit). Write $v(t) = \frac{k}{2} \log(1 + t^{-2})$, so that $F(1 + t^{-2}) = 4 \sinh^2(\frac{1}{2}v(t))$ and, with $v_H = v(H)$, $F(1 + H^{-2}) = 4 \sinh^2(\frac{1}{2}v_H)$. The normalised profile is therefore exactly a ratio of squared hyperbolic sines,

$$\frac{F(1 + t^{-2})}{F(1 + H^{-2})} = \left(\frac{\sinh(v(t)/2)}{\sinh(v_H/2)} \right)^2,$$

and since $\sinh x/x$ is increasing on $[0, \infty)$ while $\log(1 + u) \leq u$, for $t \in [H, 2H]$

$$\frac{F(1 + t^{-2})}{F(1 + H^{-2})} \leq \left(\frac{v(t)}{v_H} \right)^2 \leq (1 + \delta)^{-4} (1 + O(H^{-2})), \quad \delta = \frac{t}{H} - 1.$$

Since $\int_0^1 (1 + \delta)^{-4} d\delta = \frac{7}{24}$ exactly, and using $M(t) \leq M_{\text{up}}(t)$ from (1),

$$\text{near} \leq \frac{1}{\pi} \log(2H) \int_H^{2H} F(1 + t^{-2}) dt \leq \frac{7}{24} \cdot \frac{H \log(2H)}{\pi} \cdot F(1 + H^{-2}) (1 + O(H^{-2})).$$

The bound needs no numerical integration, and it is sharp: the true normalised profile tends to $\frac{7}{24}$ as $\lambda \rightarrow 0^+$. Over two million zeros and the grid $\lambda \in [0.5, 40]$, $H \in \{10^2, \dots, 10^5\}$, the average of the true profile is at most 0.9994 times $\frac{7}{24}$, and the computed near sum never exceeds the bound.

With the identity of Remark 6 for the right-hand factor, the comparison (4) therefore reduces to the single elementary inequality

$$\frac{7}{24} \cdot \frac{H \log(2H)}{\pi} \leq 2T(H) \quad \Longleftrightarrow \quad T(H) \geq \frac{7H \log(2H)}{48\pi},$$

which after multiplying by $48\pi/H$ becomes $\log H + \frac{64\pi a}{3} \geq 7 \log 2$, that is $\log H \geq 4.852 - 10.667 = -5.815$, true for every $H > 1$; the margin is $\log H + \frac{64\pi a}{3} - 7 \log 2 > 0$. Numerically the near bound is only about a tenth of the right-hand side at $H = 10^3$, $\lambda = 3$ (the bound gives 2.5×10^2 against 2.4×10^3), so the comparison is decided with room to spare and not by a delicate cancellation.

4 The zero sum, with its boundary term

The essential bookkeeping is the Abel identity applied to S_4 , in which the boundary term must be retained:

$$S_4(H) = \int_H^\infty t^{-4} d(2N(t)) = [2N(t)t^{-4}]_H^\infty + 8 \int_H^\infty N(t)t^{-5} dt = -2N(H)H^{-4} + 8 \int_H^\infty N(t)t^{-5} dt.$$

Substituting (1) and using $\int_H^\infty t^{-4} \log t dt = \frac{1}{3}H^{-3} \log H + \frac{1}{9}H^{-3}$ and $\int_H^\infty t^{-4} dt = \frac{1}{3}H^{-3}$ gives the following.

Lemma 5. $S_4(H) = \frac{2a}{3}H^{-3} \log H + \left(\frac{8a}{9} + \frac{2b}{3}\right)H^{-3} + \text{error}$, where $|\text{error}| \leq (4c \log H + \frac{c}{2} + 4d)H^{-4}$. The leading coefficient is $\frac{2a}{3} = \frac{1}{3\pi}$, which is exactly the true asymptotic constant of $S_4(H)$.

Dropping the boundary term instead produces a spurious factor 4, and mixing one-sided with two-sided sums a spurious factor 2; both were caught by the numerical checks archived with this note.

5 The comparison, and the exact margin

Dividing both sides by $k^2/4$ turns the comparison of Theorem 1 into

$$r_H^{(k-4)^+} S_4^+(H) \leq 2 \text{fac } T(H), \quad (4)$$

where S_4^+ is the right-hand side of Lemma 5, $\text{fac} = (r_H^k + r_H^{-k} - 2)/(k^2/4)$ is the exact Taylor factor, and $T(H) = \frac{a}{3}H \log H + \frac{4a}{9}H + 2c \log H + 2d + \frac{c}{4}$. One has $\text{fac} = u^2(1 - u + O(u^2))$ with $u = H^{-2}$.

Remark 6 (the factor on the right is the summand at the threshold). Since $r_H^k = (1 + H^{-2})^{k/2}$, the coefficient $r_H^k + r_H^{-k} - 2$ on the right of (4) is exactly the summand of Conjecture 3.2.7 of [2] evaluated at $\gamma_\rho = H$:

$$r_H^k + r_H^{-k} - 2 = (1 + H^{-2})^{k/2} + (1 + H^{-2})^{-k/2} - 2 = F(1 + H^{-2}).$$

Consequently (4) is a statement not about two unrelated expressions but about the *decay of the normalized profile* $t \mapsto F(1 + t^{-2})/F(1 + H^{-2})$ over the interval $[H, 2H]$: after dividing by the endpoint value, the tail sum is required to be at most $2T$. This is the form in which the remaining explicit estimate is most naturally attacked, because the profile is controlled by the single exponent $v(t) = \frac{k}{2} \log(1 + t^{-2})$, which is decreasing, equals $\log r_H^k$ at $t = H$, and satisfies $v(H) - v(t) = \frac{\lambda}{2}(1 - H^2 t^{-2}) + O(H^{-4})$ with $\lambda = k/H^2$.

At leading order both sides of (4) equal $\frac{2a}{3}H^{-3} \log H$; equivalently the ratio of the two sides of Theorem 1 tends to $2\pi a = 1$ exactly. The inequality is therefore decided by the subleading terms. Collecting them, the difference of the two sides is

$$\begin{aligned} \text{RHS} - \text{LHS} &= -\frac{2b}{3}H^{-3} + \left[(4c \log H + 4d + \frac{c}{2}) - (4c \log H + \frac{c}{2} + 4d) \right] H^{-4} + O(H^{-5} \log H) \\ &= \frac{2}{3}|b|H^{-3} - O(H^{-5} \log H) > 0 \quad (H > e, \text{ since } b < 0). \end{aligned}$$

So the margin is exactly $\frac{2}{3}|b|H^{-3}$: *the negative sign of the linear coefficient of the counting function is what proves the theorem*. This is consistent with, and explains, the case distinction $b \geq 0/b < 0$ in the statement of Conjecture 1.7.10 of [2].

Numerical confirmation over two million zeros ($\gamma \leq 1.132 \times 10^6$): for $H = 10^2, 10^3, 10^4$ the measured differences are 3.010330×10^{-7} , 3.011072×10^{-10} , 3.011081×10^{-13} against the predicted $\frac{2}{3}|b|H^{-3} = 3.011079 \times 10^{-7}$, 3.011079×10^{-10} , 3.011079×10^{-13} — agreement to six significant figures, and independent of k for $k \in \{2, \dots, 2048\}$.

6 Covering all (k, H) , and the two-sided bound

Both pieces of the sum now carry explicit bounds: near by Lemma 4, whose chain is the exact identity $F(1 + u) = 4 \sinh^2(\frac{1}{2}v)$, the monotonicity of $\sinh x/x$ and $\int_0^1 (1 + \delta)^{-4} d\delta = \frac{7}{24}$; and far by Lemmas 3 and 5, whose inputs are the counting hypothesis (1) only. Adding the two, and using Remark 6, the comparison with the right-hand side is established for all $k \geq 2$ and all $H > e$ with no numerical integration anywhere.

The numerical verification recorded in the companion scripts covers two million zeros and the grid $\lambda \in [0.5, 40]$, $H \in \{10^2, \dots, 10^5\}$, with no violation; the smallest margin found over that grid is $\frac{2}{3}|b|H^{-3}$ in the normalized comparison of (4), driven as explained above by the negative sign of b . For $e < H < 5$ the statement is trivial: the first zero lies at $\gamma_1 = 14.13\dots$, so the left side vanishes identically while the right side is large (at $H = 3$ the left side is $\approx 5 \times 10^{-5}$ against ≈ 0.126).

7 Honest limitations

1. Lemma 4 is now in closed form: the near-piece bound is $\frac{7}{24}H \log(2H)F(1 + H^{-2})/\pi$, obtained from the exact identity $F(1 + u) = 4 \sinh^2(\frac{1}{2}v)$, the monotonicity of $\sinh x/x$, and the elementary integral $\int_0^1 (1 + \delta)^{-4} d\delta = \frac{7}{24}$. No numerical integration enters the proof.
2. The pointwise local count implied by (1) was rejected on purpose: its leading constant is $a + 2c$, about 2.4 times the smooth density's a , and substituting it into the near bound would push the ratio in the critical band above one. The bound of Lemma 4 avoids this by using the smooth density together with the monotonicity of the profile; the Abel form recorded earlier gives the same conclusion with a ratio of at most 0.351 and served as an independent check.
3. The remaining lemmas used in the chain (partial summation over the zeros, the lower and upper bounds for the sums over the zeros beyond height $k\tau$, and the modified form of the defective lemma) are Brown's originals or Droll's generalisations; we take them at reference level and have not independently re-derived them.
4. The original article [1] is behind a paywall with no open copy. The statements of Brown's theorem and of the two errors in his lemma are therefore quoted from the thesis [2], which is the authoritative accessible source.

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References

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